Bayesian Spectral Flow Matching

Sphere benchmark, neural operators & molecular FM

ETH — Diffusion models project

bayesian-spectral-flow-matching

June 22, 2026

Outline

Target distribution on S^2

Fully synthetic ground truth (`Dataset/sphere.py`) — each sample is a random field on the sphere with known spectral law.

Pipeline (informal)

1. Draw coefficients $\boldsymbol{\xi} = (\xi_{\ell m})$ in a real spherical-harmonic basis
2. Form $f(\theta, \phi) = \sum_{\ell m} \xi_{\ell m} Y_{\ell m}(\theta, \phi)$ and evaluate on a grid
3. Vectorize f for FM training (800 cells in the main benchmark: 20×40)

Formal sampling equations on the next slide; geodesic **graph Laplacian** on the same mesh drives our spectral method.

Generative model — band-limited fields on S^2

Continuous field. For band-limit $L_{\max}$ and real SH basis $\{Y_{\ell m}\}_{0 \le \ell \le L_{\max}}$,

$$f(\theta, \phi) = \sum_{\ell=0}^{L_{\max}} \sum_{m=-\ell}^{\ell} \xi_{\ell m} \, Y_{\ell m}(\theta, \phi), \qquad \xi_{\ell m} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\big(0, \, \sigma_{\ell m}^2\big).$$

Spectral decay (smoothness knob $s > 0$, code default $s{=}2$):

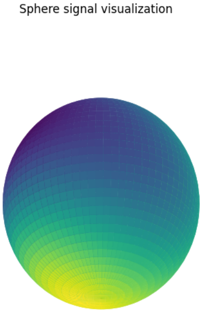
$$\sigma_{\ell m} = \begin{cases} \sqrt{\mathrm{Var}_0} & \ell = 0, \\ \big(\ell(\ell+1)\big)^{-s/2} & \ell \ge 1, \end{cases}$$

Discrete evaluation. Grid points $\{(\theta_i, \phi_j)\}$, synthesis matrix $Y \in \mathbb{R}^{N_{\mathrm{grid}} \times N_{\mathrm{coeff}}}$,

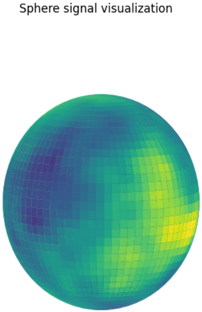
$$\mathbf{f} = Y \boldsymbol{\xi} \in \mathbb{R}^{N_{\mathrm{grid}}}, \qquad N_{\mathrm{coeff}} = \sum_{\ell=0}^{L_{\max}} (2\ell + 1).$$

Synthetic ground truth — band-limit

Random targets from `generate_coeff` → `build_synthesis_matrix`.



low band-limit $L_{\max}$=2

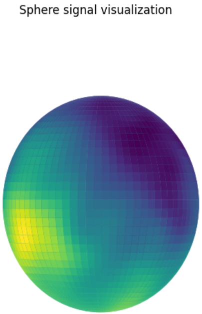


high band-limit $L_{\max}$=30

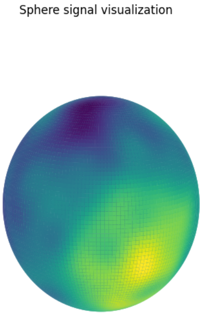
Spectral complexity (harmonic degree) — swept in later $L_{\max}$ studies.

Synthetic ground truth — mesh resolution

Same synthesis pipeline; finer $(n_\theta \times n_\phi)$ grids.



mesh 40×80



mesh 80×160

Discretization resolution at fixed L_{max}.

Geodesic graph Laplacian

Graph on the same $(n_\theta \times n_\phi)$ grid as the data: nearby cells are connected.

Edge weights from **geodesic distance** on S^2 and a **heat kernel**:

$$d_{uv} = \arccos(\mathbf{x}_u \cdot \mathbf{x}_v), \qquad w_{uv} = \exp\left(-\frac{d_{uv}^2}{2\sigma^2}\right).$$

Then $L = D - W$ (standard graph Laplacian).

Goal

Generative transport from a source distribution to band-limited random fields on the sphere, evaluated with **Wasserstein distance** on a fixed grid.

Compare:

- **Standard FM** — Gaussian source, Euclidean OT pairing, MSE flow-matching loss
- **Ours** — optional *spectral* loss weighting, *Sobolev OT* pairing on the grid Laplacian, *Matérn* source sampling

Target: coefficients in spherical harmonics up to $L_{\max}$, synthesized on an $(n_\theta \times n_\phi)$ mesh.

Three ablatable components

Flag	Module	Role
`use_spectral`	`SpectralFM`	Sobolev-weighted CFM loss in Laplacian eigenbasis (α)
`use_sobolev_ot`	`OTDisplacementMap`	Minibatch pairing under Sobolev cost (β)
`use_matern_source`	Matérn sampler	Source x_0 drawn as smooth field on graph (ν, τ)

Sphere benchmark uses geodesic grid Laplacian, SH synthesis matrix Y, and Wasserstein metric in `Tests/metrics.py`.

Experiment — setup

3-component ablation (Wasserstein evaluation on S^2)

Model & data

- MLP `intermediate_dim` = 256
- Mesh: n_θ=20, n_ϕ=40 $\Rightarrow$ 800 cells
- FM steps: 4 Epochs: 1000
- $L_{\max} = 5$

Hyperparameters

- $\alpha = -1$
- $\beta = 2.0$
- $\nu = 3.0$, $\tau = 2.0$

Experiment — Wasserstein results

Configuration	W_2 **(lower is better)**
Standard FM	25.40
Ours: spectral + Sobolev OT + Matérn (all three)	8.14
Ours: Matérn source + spectral loss	**7.98**
Ours: Matérn source only	26.14
Ours: spectral loss only	25.45

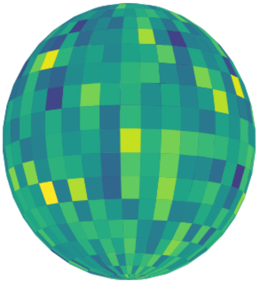
Takeaway: Enabling all three reaches $W_2 \approx 8.14$ (vs. 25.4 baseline). Best run: Matérn + spectral loss (≈ 7.98); Sobolev OT adds little on top of that pair. Single components alone stay near baseline.

Experiment — Wasserstein results — discussion

- **Matérn source is the dominant ingredient** — alone it does not help ($W_2 \approx 26$), but paired with spectral loss it drops to ≈ 8.
- **Sobolev OT is secondary** on this grid once source and loss are aligned with manifold smoothness.
- **Spectral loss alone** reweights gradients but cannot fix a mismatched Gaussian prior.

Experiment — Standard FM

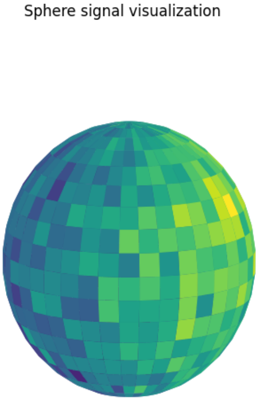
Sphere signal visualization



Gaussian source, Euclidean OT, MSE loss

$W_2 = 25.40$

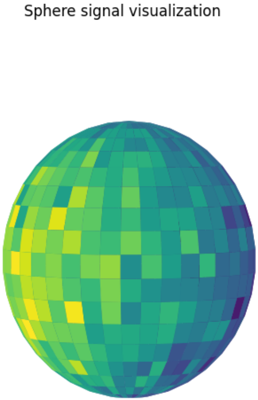
Experiment — Ours: all three enabled



Matérn source + Sobolev OT + spectral loss

$W_2 = 8.14$

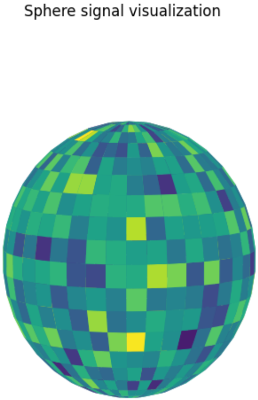
Experiment — Ours: Matérn + spectral loss



Matérn source + Sobolev-weighted CFM loss

$W_2 = \mathbf{7.98}$

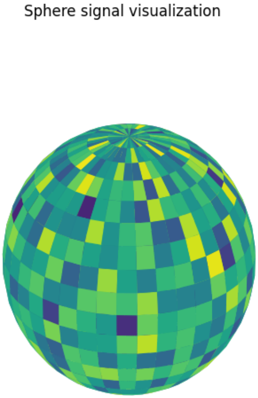
Experiment — Ours: Matérn only



Matérn source; standard loss and OT

$W_2 = 26.14$

Experiment — Ours: spectral loss only



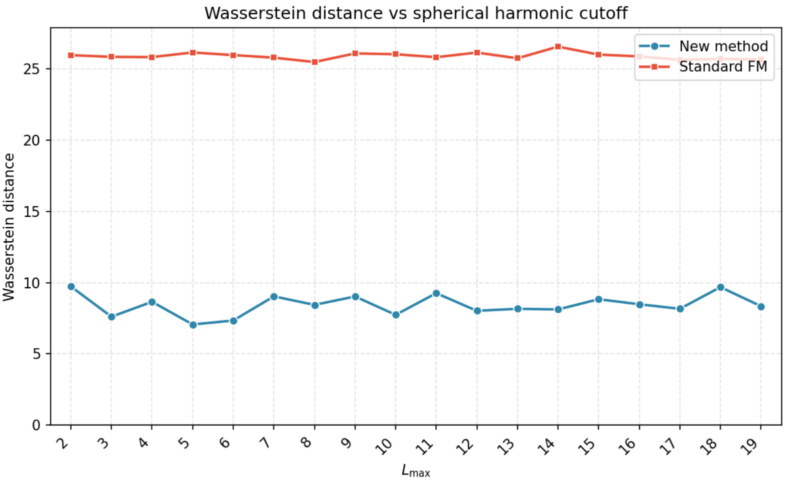
Sobolev-weighted loss; Gaussian source

$W_2 = 25.45$

Open questions

- Matérn source + spectral loss strongly outperforms Gaussian source — why?
- How should we **select / calibrate** $(\nu, \tau, \alpha, \beta)$ for a new target?
- **Bayesian update of the source** during training (adaptive prior) — promising direction?

Experiment — Wasserstein vs. L_{max}



Takeaway: Standard FM ≈ 26 (flat). Ours ≈ 7–10; minimum $W_2 \approx 7.07$ at $L_{max}=4$.

Experiment — Wasserstein vs. $L_{\max}$ — discussion

- Standard FM is **insensitive** to target band-limit — wrong inductive bias, not wrong capacity.
- Our method tracks difficulty: higher $L_{\max} \Rightarrow$ slightly higher W_2, with a sweet spot near $L_{\max}=4$.
- Non-monotonicity at 500 epochs may reflect under-convergence rather than true sensitivity.

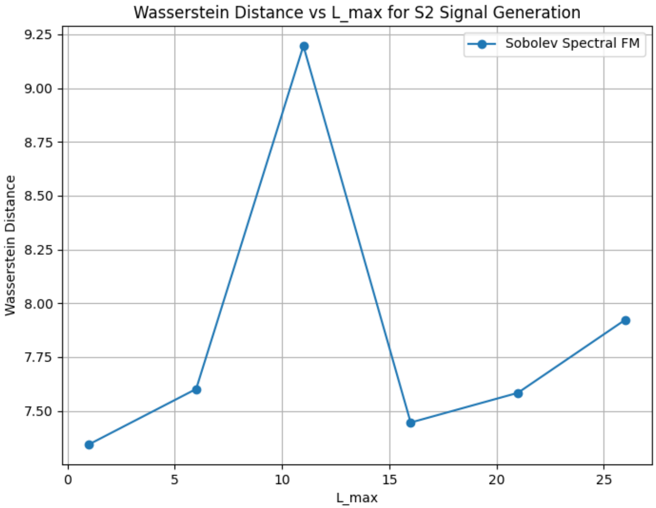
Experiment — note on convergence

This sweep used **500 epochs** (vs. 1000 in the component-ablation run).

The non-monotonic curve is likely **not the expected behavior —** probably due to insufficient convergence at 500 epochs.

Follow-up: epoch budget scaled to **1000** for the next L_{max} sweep.

Experiment — $L_{\max}$ sensitivity (1000 epochs)



Architectures

Exploring other Neural operators

- `fno2d` — standard 2D Fourier Neural Operator (FFT on the grid)
- `laplacian_fno` — graph-Laplacian eigenbasis conv (NORM-style)
- `sobolev_fno` — same basis, with Sobolev whitening in encode/decode

Neural operator in flow matching

Operator view. Treat the state as a function a on N nodes (grid cells or flattened atoms), not a single vector in $\mathbb{R}^N$:

$$v_\theta \;=\; Q \circ \mathcal{L}_T \circ \cdots \circ \mathcal{L}_1 \circ P,$$

- P (**lift**): $(a, \mathrm{coords}, t) \mapsto$ hidden channels $v \in \mathbb{R}^{C \times N}$
- $\mathcal{L}_\ell$ (**spectral layer**): integral kernel in a truncated basis + local $1{\times}1$ bypass
- Q (**project**): hidden channels $\mapsto$ per-node velocity

FM interface unchanged: `forward(x, t)` with `x` $\in \mathbb{R}^{B \times N}$, `t` $\in \mathbb{R}^{B \times 1}$. Adapters in `models/FNO.py`, `models/manifold_fno.py`.

One spectral layer — shared recipe

Every backbone layer has the same structure (FNO, Laplacian FNO, Sobolev FNO):

$$v_{\ell+1} = \sigma\Big(\underbrace{\mathcal{D}^{-1}(R_\ell \cdot \mathcal{E}(v_\ell))}_{\text{integral operator}} + \underbrace{W_\ell\, v_\ell}_{\text{local bypass}}\Big).$$

Encode $\mathcal{E}$, **decode** $\mathcal{D}^{-1}$, learned mode weights R_ℓ:

Backbone	$\mathcal{E}(v)$	R_ℓ acts on
`fno1d` / `fno2d`	$\mathrm{rfft}(v)$ / $\mathrm{rfft2}(v)$	first K Fourier modes
`laplacian_fno`	$\Phi^\top v$	first K Laplacian eigenmodes ϕ_k
`sobolev_fno`	$(1+\lambda_k)^{\alpha/2}\,\langle v, \phi_k\rangle$	whitened coefficients (decode with inverse scale)

Note on `sobolev_fno`: weighting coefficients without changing the basis $\{\phi_k\}$ (scaling only) does not buy extra expressivity—encode and decode scales cancel within each layer, so it is equivalent to `laplacian_fno`. Just a weird idea I wanted to try out.

S^2 architecture comparison — balanced capacity

Fair backbone comparison (`compare_manifold_fno_s2`): matched trainable parameter count across backbones.

Backbone	**Parameters**
`mlp`	137,120
`fno2d`	133,505
`laplacian_fno`	133,505
`sobolev_fno`	133,505

Neural operators are within $\approx 3\%$ of the MLP — differences in W_2 reflect architecture, not capacity.

Experiment — standard S^2 spherical harmonics signal

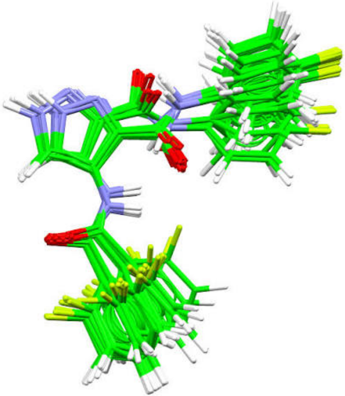
Balanced backbones ($\approx$ 137k) on band-limited SH target ($L_{\max}$=5, 20$\times$40 grid); `compare_manifold_fno_s2`.

Backbone	**Standard FM (1000 ep)**	**Spectral FM (500 ep)**	**Spectral FM (1000 ep)**
`mlp`	28.39	7.21	7.63
`fno2d`	**6.36**	13.92	7.64
`laplacian_fno`	7.02	7.62	**7.37**
`sobolev_fno`	6.40	29.15	32.44

Standard FM: Gaussian source + MSE. Spectral FM: Matérn + Laplacian spectral loss ($\alpha=-1$, $\beta=2$, $\nu=3$, $\tau=2$).

Takeaway: Injecting topological information (graph Laplacian / Fourier structure in the architecture) with standard FM reaches $W_2 \approx 6$–7, **slightly** beating our spectral FM (≈ 7.4). This SH-sampling task may be **biased** toward Fourier backbones (natural spherical basis); we need a less biased benchmark. Our spectral loss may also be **redundant or confusing** when target and backbone already share spectral geometry.

Molecular conformer generation



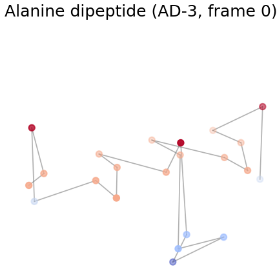
Molecular conformer generation — task

Goal: learn a generative transport map from a simple prior to **equilibrium molecular conformations** sampled by MD.

- **Input / output:** 3D atomic coordinates $x \in \mathbb{R}^{N_{\mathrm{atoms}} \times 3}$
- **Our angle:** spectral / Sobolev flow matching with geometry-aware priors (as on S^2); graph Laplacian from **covalent bonds** or **heat-kernel** on interatomic distances
- **Evaluation:** distributional match of dihedral angles, energies, and structural observables

Reference baseline: Timewarp (NeurIPS 2023) — normalizing-flow MCMC for peptide acceleration.

Alanine dipeptide — the molecule



AD-3 initial frame (`ad1-traj-state0.pdb`)

Alanine dipeptide (AD) — smallest dipeptide:

- 22 atoms (backbone + side chains)
- Standard benchmark in MD software and ML-for-chemistry

microsoft/timewarp dataset card, AD-3 folder.

AD-3 dataset (Microsoft TimeWarp)

HuggingFace `AD-3/`: `train/ad1-traj-*` (long MD) and `test/ad2-traj-*` (held-out).

Each split: `*-state0.pdb` + `*-arrays.npz` with `positions` $(T, 22, 3)$.

Training trajectory: $8{\times}10^5$ frames (OpenMM, AMBER14, implicit solvent, $T{=}310\,\mathrm{K}$).

Reported experiments subsample $\mathcal{O}(10^3)$–10^4 training frames, not the full trajectory.

Evaluation metrics

Generated $\{x_i^{\mathrm{gen}}\}_{i=1}^{M}$, reference $\{x_j^{\mathrm{true}}\}_{j=1}^{M}$ (COM-relative coords, $x \in \mathbb{R}^{66}$).

Euclidean W_2

$$d^2_{\mathrm{Eucl}}(x, y) = \|x - y\|_2^2, \quad W_2 = \sqrt{\min_{\pi \in \Pi(a,b)} \sum_{i,j} \pi_{ij}\, d^2_{\mathrm{Eucl}}(x_i^{\mathrm{gen}}, x_j^{\mathrm{true}})},$$

uniform weights $a_i = b_j = 1/M$; exact EMD (`ot.emd2`).

Sobolev W_2 (ϕ_k, λ_k from bond-graph / heat-kernel Laplacian; β=2 on AD-3)

$$d^2_{\mathrm{Sob}}(x, y) = \sum_{k=1}^{K} (1 + \lambda_k)^{-\beta} \langle \phi_k, x - y \rangle^2, \quad W_{2,\mathrm{Sob}} = \sqrt{\min_{\pi} \sum_{i,j} \pi_{ij}\, d^2_{\mathrm{Sob}}(x_i^{\mathrm{gen}}, x_j^{\mathrm{true}})}.$$

COM-aligned RMSD reshape $x \to \{r_a\}_{a=1}^{22} \subset \mathbb{R}^3$, $\tilde{r}_a = r_a - \bar{r}$, $\bar{r} = \frac{1}{22} \sum_a r_a$,

$$\mathrm{RMSD}(g, t) = \sqrt{\frac{1}{22} \sum_{a=1}^{22} \|\tilde{g}_a - \tilde{t}_a\|_2^2}, \quad \overline{\mathrm{RMSD}} = \frac{1}{M} \sum_{i=1}^{M} \min_j \mathrm{RMSD}(g_i, t_j) \ \text{(nm)}.$$

Per-atom structural error after center-of-mass alignment; each generated conformer is matched to its closest reference. `Tests/metrics.py` Lower is better for all three.

Experiment — setup

Conformer generation on AD-3 (alanine dipeptide, Microsoft TimeWarp)

Data & representation

- Train: `ad1-traj` (8×10^5 frames, subsample 8192)
- Test: `ad2-traj` (1024 held-out frames; final comparison also uses 2048)
- State: COM-relative coords, $22\times3=66$ dims
- Graph: covalent bond Laplacian (`molecule_graph.py`)

Training

- MLP or balanced neural-operator backbones ($\approx$ 137k params)
- FM steps: 8 Epochs: 1000 Batch: 512
- $\alpha{=}-1$, $\beta{=}2.0$, $\nu{=}3.0$, $\tau{=}2.0$
- Ours: Matérn source + Laplacian spectral loss

Experiment — standard vs. spectral FM (MLP)

Same MLP backbone (`intermediate_dim=512`); only FM recipe differs.

Configuration	W_2 **(Eucl.)**	W_2 **(Sob.)**	**RMSD mean (nm)**
Standard FM (Gaussian source, MSE loss)	0.868	—	0.172
Ours (Matérn source + spectral loss)	**0.671**	**0.522**	**0.126**

Takeaway: On real MD conformations, the S^2 recipe transfers — spectral FM cuts Euclidean W_2 by $\approx 23\%$ and RMSD by $\approx 27\%$ vs. standard FM.

Experiment — standard vs. spectral FM — discussion

- **Transfer to real data:** Matérn prior + spectral loss helps beyond synthetic S^2 fields.
- Sobolev W_2 (0.52) is lower than Euclidean (0.67) — metric aligns with smooth molecular graphs.
- COM-aligned RMSD $\approx 0.13\,\text{nm}$ is chemically reasonable for a small peptide scaffold.

Experiment — architecture comparison (balanced capacity)

Fair comparison at $\approx$ 137k trainable parameters per backbone; FM recipe varies by column.

	Standard FM		Spectral FM (ours)		
Backbone	W_2	**RMSD**	W_2	W_2^{Sob}	**RMSD**
`fno1d`	1.340	**0.167**	**0.753**	**0.624**	**0.114**
`mlp`	**1.015**	0.206	0.815	0.662	0.146
`graph_laplacian_fno`	1.382	0.188	0.944	0.751	0.159
`graph_sobolev_fno`	1.336	0.177	1.202	0.986	0.185

Standard FM: Gaussian source + MSE loss. Spectral FM: Matérn source + Laplacian spectral loss.

Takeaway: Spectral FM improves every backbone ($W_2 \downarrow$ 25–44%, RMSD $\downarrow$ 18–32%); `fno1d` best overall under our recipe. *Likely reason: `fno1d` works in Fourier space, where convolution is diagonal—a natural match for learning spectrally structured velocity fields.*

Different graph Laplacians representing the molecule

Experiment — heat-kernel graph Laplacian

Motivation: replace covalent bonds with a **geometry-aware** graph from 3D distances.

Edge weights from Euclidean interatomic distance (reference PDB frame):

$$w_{ij} = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right), \qquad L = D - W,$$

with σ defaulting to the **median pair distance** among included atoms (`graph_type="heat"` in `molecule_graph.py`).

Same spectral FM recipe (Matérn source + Laplacian spectral loss); compare `graph_type="heat"` vs. `"bond"` at matched 1000-epoch budget (final results on next slide).

Experiment — heat vs. bond graph (final comparison)

Matched setup: 1000 epochs, 16384 train frames, batch 1024, 2048 test, 8 FM steps.

	Spectral FM (heat)			**Spectral FM (bond)**			**Standard FM (bond)**		**Standard FM (heat)**	
Backbone	W_2	RMSD	W_2^{Sob}	W_2	RMSD	W_2^{Sob}	W_2	RMSD	W_2	RMSD
`mlp`	1.24	0.236	0.14	0.72	0.139	**0.57**	**0.90**	**0.184**	**0.90**	0.185
`fno1d`	**1.08**	**0.117**	0.38	<u>**0.70**</u>	<u>**0.107**</u>	0.58	1.27	0.155	1.29	**0.158**
`graph_laplacian_fno`	1.21	0.203	**0.11**	1.12	0.197	0.88	1.38	0.193	1.29	0.173
`graph_sobolev_fno`	1.23	0.189	0.12	0.95	0.151	0.76	1.36	0.186	1.28	0.168

Spectral FM: Matérn source + Laplacian spectral loss. Standard FM: Gaussian source + MSE. Bold = best per column; underline = best overall Euclidean W_2 / RMSD. For Standard FM, bond vs. heat only enters via the Laplacian in `graph_laplacian_fno` / `graph_sobolev_fno` (`mlp`, `fno1d` unchanged).

Takeaway: Bond graph wins on Euclidean W_2/RMSD under spectral FM (`fno1d` best: W_2=0.70, RMSD 0.11 nm). Heat graph yields very low Sobolev W_2 ($\approx$ 0.11–0.14) but worse Cartesian fit.

Experiment — discussion & open questions

- **Confirmed:** Matérn + spectral loss helps on real molecular data, not only synthetic S^2 fields.
- **Best backbone so far:** `fno1d` (RMSD $\approx 0.11\,\mathrm{nm}$); MLP competitive.
- **Still open:** Ramachandran / basin-occupancy histograms vs. MD (not yet plotted).

Summary & next steps

- Synthetic S^2 fields provide a controlled sanity check before harder real data.
- Component ablation: Matérn + spectral loss best ($W_2 \approx 7.98$); Matérn source is the dominant ingredient.
- Standard S^2 SH signal: under standard FM, `fno2d` best ($W_2 \approx 6.4$); under spectral FM, all except `sobolev_fno` ≈ 7.4 at 1000 epochs.
- AD-3 conformers: spectral FM beats standard (W_2 0.67 vs. 0.87; RMSD 0.13 vs. 0.17 nm); `fno1d` best among balanced backbones.
- Heat vs. bond graph (1000 epochs): bond + spectral FM best on Euclidean W_2/RMSD (`fno1d` 0.70 / 0.11 nm); heat graph yields Sobolev $W_2 \approx 0.11$–0.14 at the cost of Cartesian fit.
- Still open: Ramachandran evaluation on AD-3.

Theory — spectral setup

On the grid Laplacian L with eigenpairs (λ_k, ϕ_k), our method lives in a **Sobolev-weighted** space:

$$\|f\|_{H^\alpha}^2 = \sum_k (1+\lambda_k)^\alpha \, |\langle f, \phi_k \rangle|^2.$$

Three ingredients (component ablation):

- **Matérn source** — prior at $t{=}0$: $x_0 \sim \mathcal{N}\big(0, (\tau I + L)^{-\nu}\big)$
- **Sobolev (spectral) loss** — CFM residual weighted by $(1+\lambda_k)^\alpha$ in `SpectralFM`
- **Target** — band-limited SH field with $\mathrm{Var}(\xi_{\ell m}) \propto \big(\ell(\ell{+}1)\big)^{-s/2}$

Empirical fact: Matérn alone ($W_2 \approx 26$) and spectral loss alone ($W_2 \approx 25$) match standard FM; **both together** ≈ 7.98.

Theory — Matérn source alone fails

Matérn fixes **where you start**; standard L^2 loss trains in the **wrong metric**.

1. Prior is correct, objective is not. $\mathrm{Var}(\langle x_0, \phi_k \rangle) = (\tau + \lambda_k)^{-\nu}$, but CFM minimizes

$$\mathcal{L} = \mathbb{E}\, \| v_\theta(t, x_t) - v^\star \|^2_{\mathbb{R}^d}$$

over **grid coordinates**, not Laplacian modes. High-frequency mesh directions dominate the L^2 norm of the velocity residual even when x_0, x_1 are smooth.

2. Transport stays hard in ambient $\mathbb{R}^d$. Matérn μ_0 and smooth target μ_1 are close in H^α, but L^2 training optimizes a flat Euclidean flow. Correct boundary condition, wrong inner product — the learned ODE does not converge to the geometrically correct map.

Matérn $\Rightarrow \mu_0 \in \mathcal{H}^s$; L^2 CFM optimizes in L^2(grid) — different topologies.

Theory — Sobolev loss alone fails

Sobolev loss fixes **how you train**; Gaussian source leaves **where you start** wrong.

1. White source vs. smooth target. $x_0 \sim \mathcal{N}(0, I)$ gives $\mathrm{Var}(\langle x_0, \phi_k \rangle) = 1$ for all k, while target variance decays as λ_k^{-s}. For large λ_k, white noise has orders of magnitude more high-frequency energy than the target, so

$$x_1 - x_0 \approx x_1 - \text{(large HF noise)}.$$

2. Reweighting $\neq$ fixing the initial measure. With $\alpha = -1$, the loss downweights high λ_k, but at inference we still integrate from fresh $\mathcal{N}(0, I)$ samples. The marginal pushforward at $t{=}0$ remains the wrong measure on the manifold.

3. RKHS view. Any map from isotropic Gaussian to a smooth GP must create large Cameron–Martin norm (highly oscillatory trajectories). Downweighting those modes in the loss does not make white noise a good prior.

Theory — why both together work

The two components are **dual** with respect to the same operator L:

$$\boxed{\text{Prior: } C_0 = (\tau I + L)^{-\nu} \quad \longleftrightarrow \quad \text{Metric: } \|\cdot\|_{H^\alpha}, \quad \alpha < 0}$$

With $\nu{=}3$, $\tau{=}2$, $\alpha{=}-1$, $s{=}2$ (component ablation):

- **Aligned endpoints** — x_0 and x_1 are smooth fields with comparable spectral decay; $x_1 - x_0$ is a small structured perturbation in H^α, not "smooth minus white noise."
- **Matched geometry** — Sobolev loss trains the vector field in the same norm where μ_0 and μ_1 are close (approx. Sobolev OT between compatible GPs).
- **Band-diagonal CFM** — low λ_k modes carry transport; high λ_k modes have tiny variance *and* tiny loss weight.

Kernel-methods principle: prior and regularizer must share the same kernel — both built from L.

Theory — Bayesian reading & corollaries

Bayesian view.

- Matérn source = **prior** on the function space
- Sobolev loss = **likelihood / regularization** in the same space
- Using only one is inference with the right prior but wrong likelihood, or vice versa

Why Sobolev OT adds little (7.98 vs. 8.14): once source and loss are aligned, random (x_0, x_1) pairs are already reasonable in H^β; OT mainly helps when endpoints are spectrally mismatched.

Prior calibration: choose (ν, τ) so that $(\tau + \lambda_k)^{-\nu} \approx \lambda_k^{-s}$ on modes that carry target mass — (ν, τ, α) are one geometric specification, not independent knobs.

One-line summary: Matérn aligns the initial measure with the manifold; Sobolev loss aligns the training metric with the same manifold. Either fix alone leaves a prior–metric mismatch; both together make CFM approximate Sobolev transport between two compatible Gaussian measures.